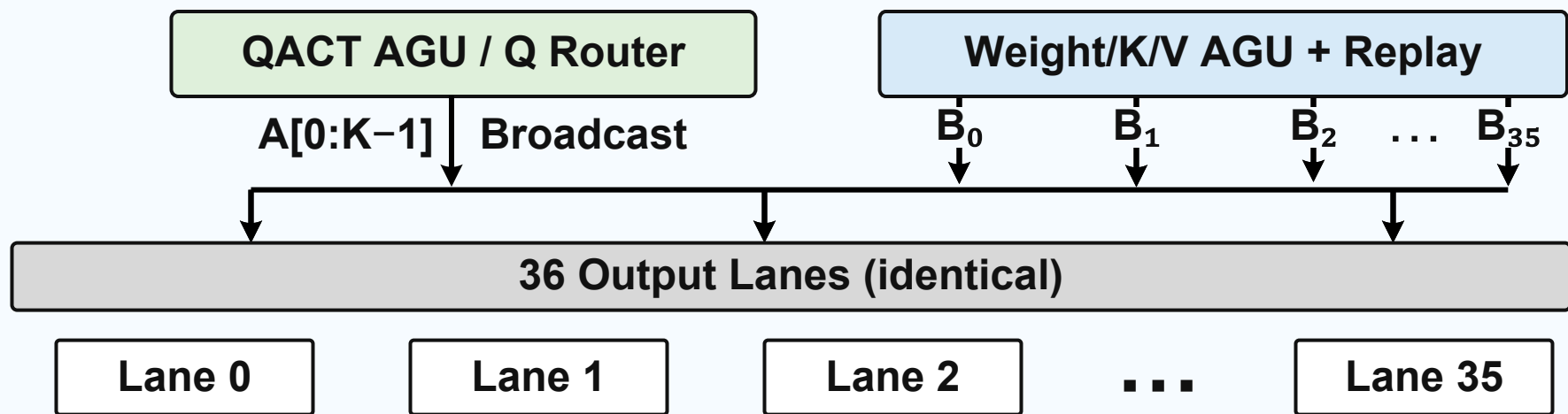
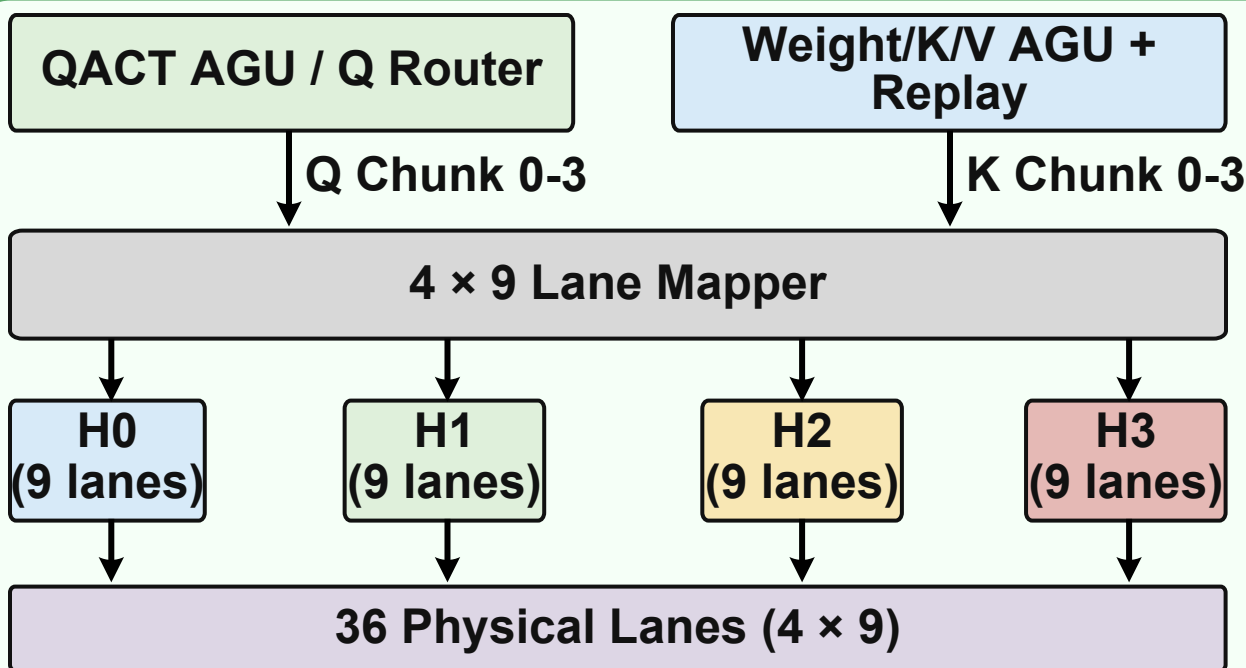




- Output dim. (N) across 36 lanes
- Reduction dim. (K) within each lane

(a) Standard spatial/linear mode (conventional GEMM-style mapping)



Key-dimension tiling (example: $T = 19$)

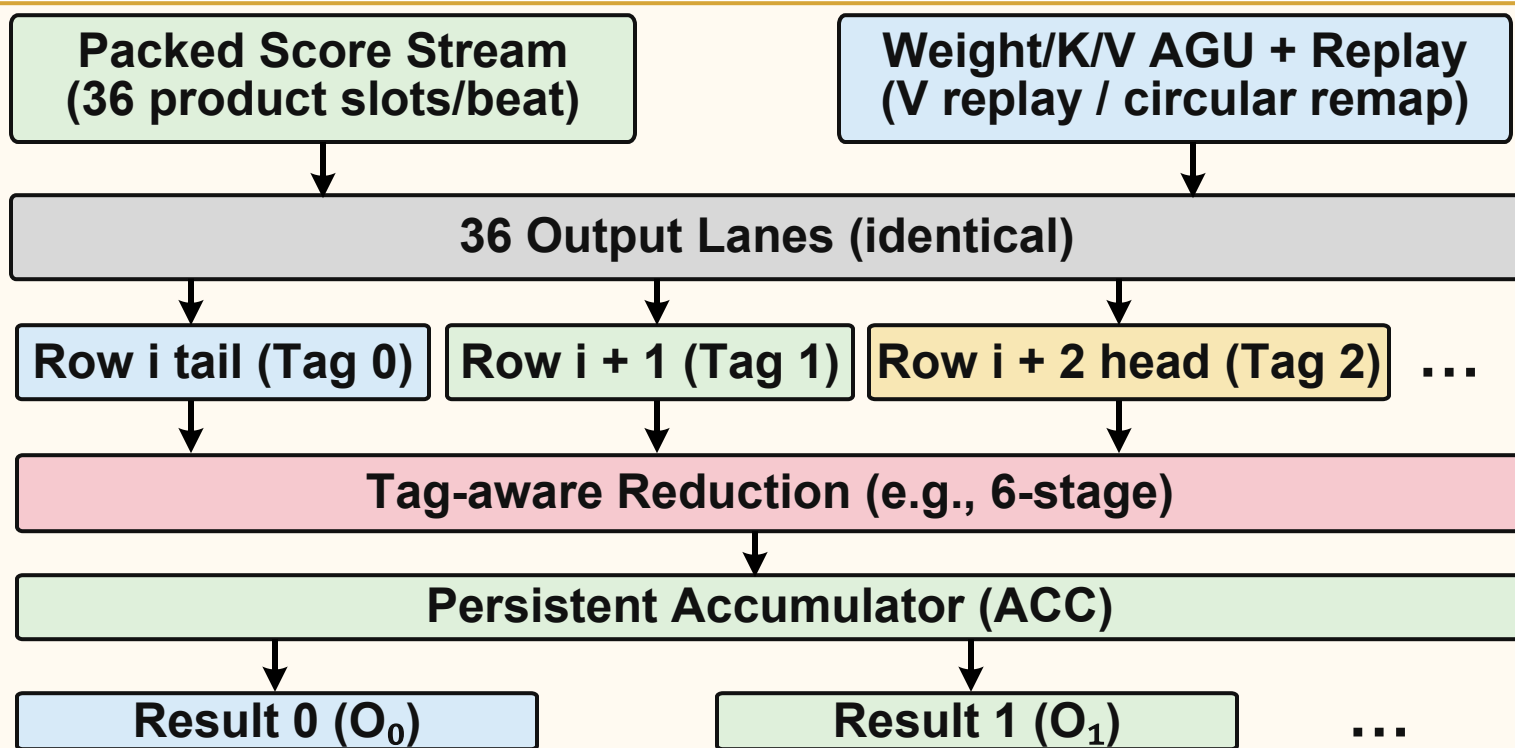
	H0	H1	H2	H3
Tile 0	0–8	0–8	0–8	0–8
Tile 1	9–17	9–17	9–17	9–17
Tile 2	18	18	18	18

Lane utilization: $19/36 = 52.8\% \rightarrow (4 \times 19)/(3 \times 36) = 70.4\%$

Overlapped Q/K delivery

Cycle	0	1	2	3	...
LOAD	0-3	4-7	8-11	12-15	
Compute		0-3	4-7	8-11	12-15

(b) Temporal QK with output-axis adaptation (4-head packing with overlapped Q/K delivery)



Segment handoff

- Tail + ACC $\rightarrow$ Output (O_0)
- Complete $\rightarrow$ Output (O_1)
- Head $\rightarrow$ next ACC

- Product-slot utilization:
- $52.8\% \rightarrow 91.2\%$ (packing + replay)

(c) Temporal PV with reduction-axis adaptation (V replay + segmented reduction)